

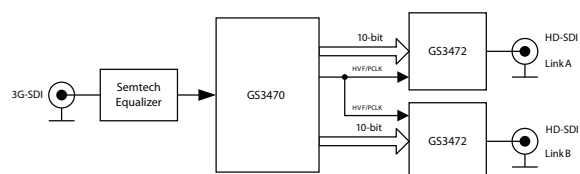
Gennum Products

Key Features

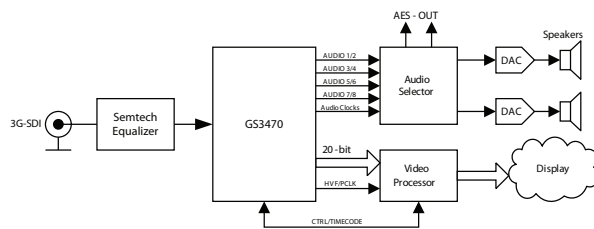
- Operation at 2.970Gb/s, 2.970/1.001Gb/s, 1.485Gb/s, 1.485/1.001Gb/s and 270Mb/s
- Supports SMPTE ST 425 (Level A and Level B), SMPTE ST 424, SMPTE 292, SMPTE ST 259-C and DVB-ASI
- Integrated Reclocker with serial digital reclocked, or non-reclocked loop-through output
- Integrated low phase noise VCO
- Integrated audio de-embedder for 8 channels of 48kHz audio and audio clock generation
- Ancillary data extraction
- Parallel data bus selectable as either 20-bit or 10-bit
- Comprehensive error detection and correction features
- Output H, V, F or CEA 861 timing signals
- GSPI host interface
- +1.2V digital core power supply, +1.2V and +2.5V analog power supplies, and selectable +1.8V or +2.5V I/O power supply
- -20°C to +85°C operating temperature range
- Low power operation – typically 220mW
- Small 9mm x 9mm 100-ball BGA package (0.80mm Ball Pitch)
- Pb-free and RoHS compliant

Applications

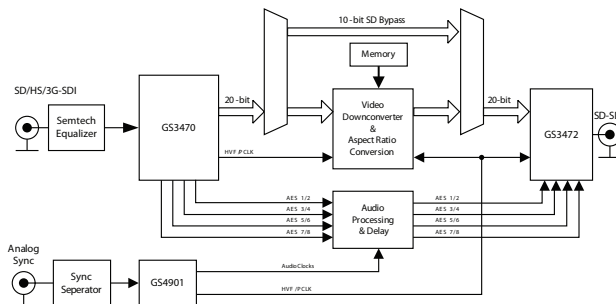
Application: Single Link (3G-SDI) to Dual Link (HD-SDI) Converter



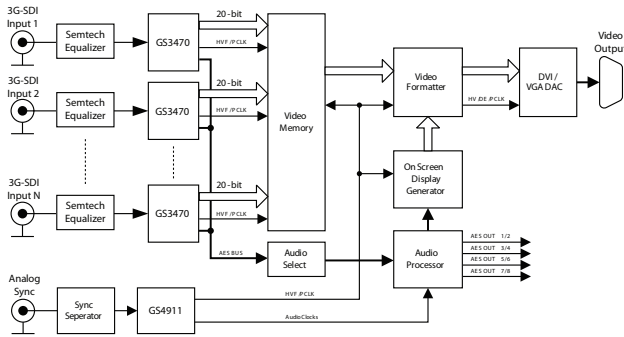
Application: 1080p50/60 Monitor



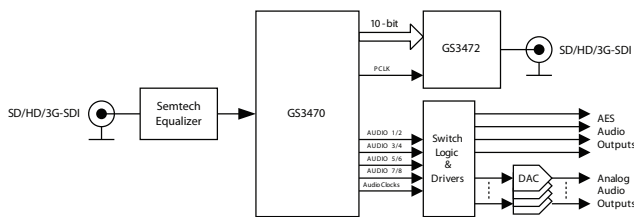
Application: Multi-format Downconverter



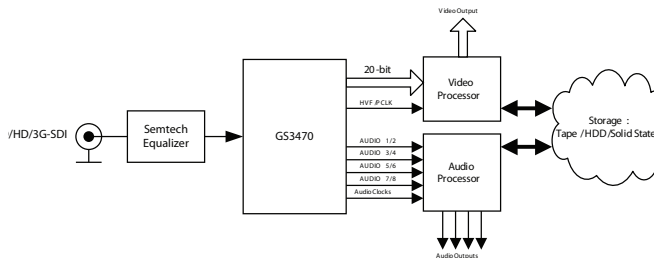
Application: Multi-input Video Monitoring System



Application: Multi-format Audio De-embedder Module



Application: Multi-format Digital VTR/Video Server



Description

The GS3470 is a multi-rate SDI Receiver which includes complete SMPTE processing, as per SMPTE ST 425, SMPTE ST 292 and SMPTE ST 259-C. The SMPTE processing features can be bypassed to support signals with other coding schemes.

The device features an Integrated Reclocker with an internal VCO and a wide Input Jitter Tolerance (IJT) of 0.7UI.

A serial digital loop-through output is provided, which can be configured to output either reclocked or non-reclocked serial digital data. The serial digital output can be connected to an external cable driver.

The device operates in one of four basic modes: SMPTE mode, DVB-ASI mode, Data-Through mode or Standby mode.

In SMPTE mode (the default operating mode), the GS3470 performs full SMPTE processing, and features a number of data integrity checks and measurement capabilities.

The device also supports ancillary data extraction, and can provide entire ancillary data packets through host-accessible registers. It also provides a variety of other packet detection and error handling features. All of these processing features are optional, and may be individually enabled or disabled through register programming.

Both SMPTE ST 425 Level A and Level B inputs are supported.

In DVB-ASI mode, sync word detection, alignment and 8b/10b decoding is applied to the received data stream.

In Data-Through mode all forms of SMPTE and DVB-ASI processing are disabled, and the device can be used as a simple serial to parallel converter.

The device can also operate in a lower power Standby mode. In this mode, no signal processing is carried out and the parallel output is held static.

Parallel data outputs are provided in 20-bit or 10-bit format for 3Gb/s, HD and SD video rates, with a variety of mapping options. As such, this parallel bus can interface directly with video processor ICs, and output data can be multiplexed onto 10 bits for a low pin count interface.

Up to eight channels (two audio groups) of serial digital audio may be extracted from the video data stream, in accordance with SMPTE ST 272-C and SMPTE ST 299.

The output audio formats supported by the device include AES/EBU and I²S, and two other industry standard serial digital formats. A variety of audio processing features are provided to ease implementation. Audio clocks are internally generated and provided by the device.



DOCUMENT IDENTIFICATION
PRODUCT BRIEF

The product is in a development phase and specifications are subject to change without notice. Semtech reserves the right to remove the product at any time. Listing the product does not constitute an offer for sale.

CAUTION

ELECTROSTATIC SENSITIVE DEVICES
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